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Plasma B-Endorphin Levels in Patients With Self-Injurious Behavior and Stereotypy

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*B-endorphin and cortisol concentrations were examined in the plasma of mentally retarded patients who displayed symptoms of self-injurious behavior (SIB) (*n = 9*), stereotypy (*n = 17*), or SIB plus stereotypy (*n = 14*). Morning and evening samples were compared with a matched patient control group (*n = 13*) and a group of nonretarded controls (*n = 17*). Compared with nonretarded controls, the retarded control group had significantly lower b-endorphin concentrations for the morning sample and nonsignificantly lower concentrations for the evening sample. The combined patient groups had lower b-endorphin levels than did nonretarded controls for both samples. Patients with SIB and stereotypy had significantly higher morning levels of b-endorphin than the retarded controls. The evening levels of the SIB plus stereotypy group also were higher than retarded controls but were not significant. Levels in the SIB and stereotypy group were slightly elevated compared with retarded controls. Cortisol levels were identical for all groups. B-endorphin and cortisol significantly covaried for the nonretarded controls but not the patients. Results indicated that compared to a matched control group, patients with SIB plus stereotypy have elevated b-endorphin plasma.*

The endogenous opiates comprise a family of neuropeptides including the enkephalins, endorphins, and dynorphin. Perhaps the most interesting of these is b-endorphin because of its many behavioral effects, including analgesic and euphoro-

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genic effects (Bloom, Segal, Ling, & Guillemin, 1976; Jacquet & Marks, 1976; Sandman, Kastin, & Schally, 1981; Sandman & O'Halloran, 1986). It is derived from a larger molecule (pro-opiomelanocortin—POMC) containing other neuropeptides (e.g., MSH, ACTH), which improve attention and memory of rats (Sandman, Kastin, Miller, & Denman, 1972), nonretarded persons (Kastin et al., 1971; Sandman et al., 1977; Ward, Sandman, George, & Shulman, 1979), and retarded individuals (Sandman, George, Walker, Nolan, & Kastin, 1976; Sandman, Walker, & Lawton, 1980; Walker & Sandman, 1979). Exactly how POMC fragments influence the brain is unknown. It is synthesized intracellularly via ribosomal mRNA translation, in the endoplasmic reticulum and packaged for release by the Golgi apparatus. Converting enzymes in the Golgi cleave POMC at amino acid base pairs (e.g., lysine and arginine), liberating the bioactive chains. The major brain site of origin for b-endorphin cell bodies appears to be the arcuate nucleus, but processes from the cell bodies have been mapped throughout the brain (Watson & Akil, 1981).

Recently, the endogenous opiate system has been proposed as a possible biological substrate of autistic behavior (Deutsch, 1986; Panksepp, 1979; Panksepp, Conner, Forister, Bishop, & Scott, 1983) and, specifically, self-injurious behavior—SIB (Davidson, Kleene, Carroll, & Rockowitz, 1983; Farber, 1987; Gillberg, Terenius, & Lonnerholm, 1985; Richardson & Zaleski, 1983; Sandman, Barron, Crinella, & Donnelly, 1987; Sandman, Datta, Barron, Hoehler, Williams, & Swanson, 1983; Sandyk, 1985). However, physiological measures of endogenous opiates have been confusing. Weizman, Weizman, Tyano, Szekely, and Sarne (1984) reported that "humoral" endorphin, a unique endogenous opiate, was decreased in plasma of neuroleptically treated autistic patients compared to nonautistic controls. This conflicted with the animal-based speculations of Panksepp and his colleagues, who argued that autistic patients were addicted to elevated infantile levels of b-endorphin (Panksepp, 1979; Panksepp et al., 1983). Gillberg et al., 1985, studying b-endorphin levels in cerebrospinal fluid failed to find either a hypo- or hyperactive opiate hypothesis in autism. However, the autistic children with SIB had significantly higher cerebrospinal fluid b-endorphin levels than did patients who did not exhibit SIB.

Perhaps the strongest evidence linking SIB with the opiate system is treatment with opiate antagonists. Several investigators have demonstrated attenuation or elimination of SIB with naloxone (Davidson et al., 1983; Richardson & Zaleski, 1983; Sandman et al., 1983; Sandman et al., 1987; Sandyk, 1985) and naltrexone (Campbell, Adams, Small, Tesch, & Currens, 1988; Herman et al., 1987). The current study was designed to examine further the relation between the endogenous opiate system and SIB by testing plasma levels of b-endorphin in patients with and without SIB. In addition, plasma cortisol was measured to evaluate the specificity of b-endorphin in these behaviors.

Method

Subjects

Subjects ranging in age from 19 to 39 from Fairview Developmental Center in Costa Mesa were separated into four groups (Table 1), based on previously described criteria (Barron & Sandman, 1983, 1985). The groups included patients with (a) SIB and stereotypy (n = 14), (b) SIB only (n = 9), (c) stereotypy only (n = 17), and (d) a retarded control group (n = 13) matched with the same diagnostic groups who did not display either stereotypy or SIB. Plasma was also collected from an age-matched external control group of normally functioning subjects (n = 17) whose characteristics closely matched the patient groups (Table 1).

Definitions

Self-Injurious Behavior. Patients evidencing repetitious self-directed behavior resulting in relatively immediate physical harm and/or tissue damage were placed in the SIB

Table 1
Characteristics of Controls and Patients

Characteristic	Nonretarded controls	Retarded patients SIB	ST[a]	SIB + ST	Controls
Sex					
Male	9	8	9	6	8
Female	8	1	8	8	5
Mean CA[b]	29.1	24.7	25.0	23.8	24.3
	(10.8)	(2.8)	(2.5)	(4.3)	(6.2)
Paradoxical					
Responders	—[c]	2	5	4	2
Nonresponders	—	7	12	10	11

[a] Stereotypy. [b] *SD* in parentheses. [c] Not applicable.

group. These behaviors were exhibited in a wide variety of ways (e.g., self-striking, self-biting, and/or tissue damage) (see Barron & Sandman, 1984).

Stereotyped Behavior. Patients displaying fixed, highly repetitive, invariant motor sequences or acts (e.g., body-rocking, head-rolling, and hand-waving) that seemed to have no apparent function and were not injurious to the individual were placed in the stereotypy group.

Materials and Procedure

Blood samples were collected for b-endorphin measurements at 8 AM corresponding to the diurnal maxima of opiate secretion (Naber et al., 1981) and at 8 PM at the diurnal trough. Blood (10 ml) was withdrawn by antecubital venipuncture into EDTA (purple top) vacutainers, with the plasma separated by centrifugation (2000 xg; 15 minutes) and aspirated into polypropylene tubes containing 500 KIU/ml aprotinin (Sigma Chemical Co., St. Louis, MO). The plasma samples were stored at −70° C until assayed.

B-Endorphin Assay

Plasma levels of b-endorphin were determined by a solid phase two-site immunoradiometric assay (IRMA) using the Allegro Immunoassay System (Nichols Institute Diagnostics, San Juan Capistrano, CA). The antiserum employed has 1.6% cross-reactivity for beta-lipotropin at 500 pg/ml and has less than 0.01% cross-reactivity with related opiates at 5 ug/ml. Samples were assayed in duplicate (200 ul per assay tube); ^{125}I-labeled anti-b-endorphin (rabbit) solution (100 ul) was added to each tube and vortexed. The reaction was initiated by adding one anti-b-endorphin (rabbit) coated polystyrene bead to the assay tube followed by a stationary incubation at room temperature for 20±4 hours. The beads were then washed twice with phosphate buffered saline and aspirated to dryness. The labeled antibody complex bound to the solid phase was measured using a Micromedic Isoflex Gamma Counter. Results were quantified by the 4-parameter logistic method using the "two plus two" iterative procedure with re-weighting developed by Rodbard at the National Institutes of Health (Program RIA004, Biomedical Computing Technology Information Center, Nashville, TN). The Allegro Beta-Endorphin Immunoassay system has a minimum detectable dose of 10 pg/ml (95.0% confidence limit) with a coefficient of variance of 4.1% (intraassay) and 9.0% (interassay) at the highest concentration measured in the present study.

Cortisol Assay

Plasma cortisol levels were assayed by immunofluorescence using an automated procedure on an Abbott TDx Analyzer (Abbott Laboratories, Abbott Park, IL). The sensitivity of the TDx Cortisol Assay is minimum detectable dose of 0.45 ug/dL with a coefficient of variance of 8.42% (intraassay) and 7.23% (interassay) at 4 ug/dL and 2.65% (intraassay) and 2.7% (interassay) at 40 ug/ml. Antiserum cross-reactivity is less than 5.0% for endogenous steroids and less than

1.2% for synthetic steroids at 1000 ug/dL (except prednisone, which has 23.8% cross-reactivity at 100 ug/dL).

Results

Comparison of Control Groups

Plasma concentration of b-endorphin for both morning, $t(18.5) = 4.13, p < .001$, and evening, $t(20.2) = 3.45, p < .002$, samples were significantly lower for retarded controls than nonretarded controls (see Figure 1). Cortisol concentrations were not significantly different for the groups at either sampling period (Figure 1). The b-endorphin: cortisol ratio was significantly different for the morning, $t(18.5) = 3.52, p < .01$, but not the evening sample. This effect was due to the large drop in evening b-endorphin concentration in nonretarded controls. The cortisol rhythm was identical in both groups.

Comparison With Nonretarded Controls and Patients Exhibiting SIB and Stereotypy

Similar results were obtained when nonretarded controls were compared with all patients exhibiting stereotypy or SIB (Figure 1). The patient groups had significantly lower morning and evening plasma b-endorphin concentration levels than did nonretarded controls, ts (3.14 and 21.2) = 3.14 and 2.90, respectively, $p\text{s} < .01$. Plasma cortisol concentration did not differentiate the groups. The b-endorphin:cortisol ratio was significantly lower in the patients for the morning sampling period, $t(30.5) = 2.72, p < .01$.

Comparison of Retarded Controls and Patients Exhibiting SIB and Stereotypy

Concentration of plasma b-endorphin in the morning sample was significantly elevated in groups with SIB or stereotypy compared with the retarded controls, $t(45) = 2.12, p < .05$; Figure 1). The b-endorphin concentration also was higher in the patients for the evening sample, but the difference was not statistically significant. Neither plasma cortisol concentration nor the b-endorphin: cortisol ratio were significantly different in the retarded controls compared with the patients evidencing stereotypy and/or SIB.

Comparison of each patient group with the retarded control group is presented in Figure 2. The SIB plus stereotypy group had higher b-endorphin morning concentrations

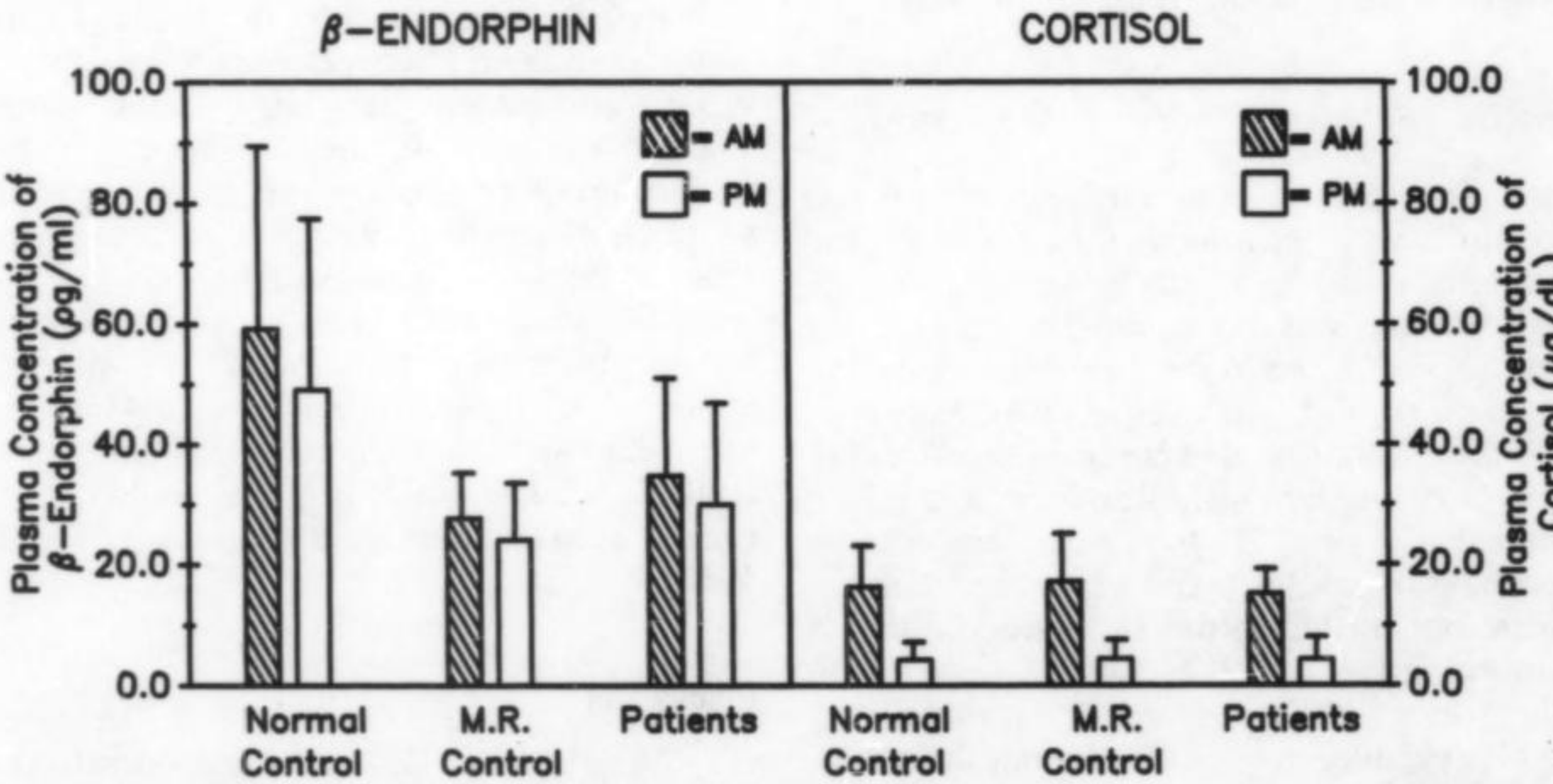

Figure 1. Concentrations of b-endorphin and cortisol for control subjects and all patients combined for morning and evening measurements.

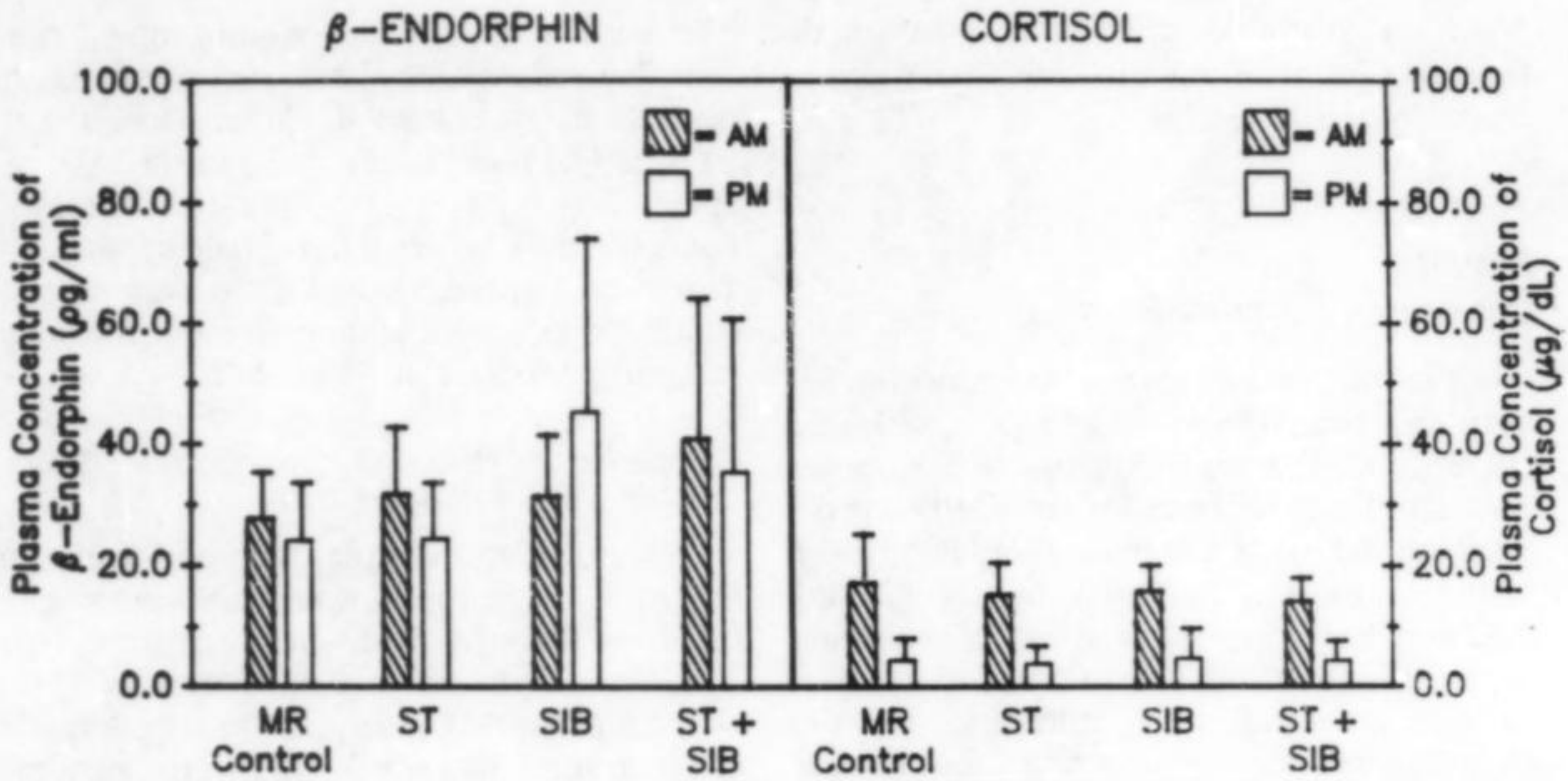

Figure 2. Concentrations of b-endorphin and cortisol in the morning and evening. ST = stereotypy, SIB = self-injurious behavior.

than ~~did~~ controls, but the effect was only marginally significant, $t(15.9) = 2.00, p < .06$. Although the other patient groups had higher b-endorphin concentrations than did retarded controls, the differences failed to achieve statistical significance. Similarly, all of the patient groups had higher evening b-endorphin levels than did controls, but the effects were not statistically significant.

Relation Between B-Endorphin and Cortisol

The relation between b-endorphin and cortisol was examined for morning and evening values. For all 79 subjects considered, there was no significant relation for samples collected in the morning or evening, rs = $-.01$ and $-.04$, respectively. However, for the 17 nonretarded controls, b-endorphin and cortisol were significantly related in the morning, $r = .51$, $p < .05$, but not the evening, $r = .34$; b-endorphin and cortisol were not related in the 13 retarded controls (morning $r = -.12$; evening $r = .10$) or in the patient group ($n = 37$; morning $r = -.15$; evening $r = -.09$). Identical findings were obtained when the patient groups were analyzed separately.

Confounding Variables

There were no significant relations among age, sex, and levels of b-endorphin and cortisol. Because medication can influence plasma b-endorphin concentrations, patients were divided into groups on the basis of their taking analgesics, antihistamines, anticonvulsants, antipsychotics, or sedatives. Concentrations of b-endorphin were compared between each of these groups and patients not taking medication. There were no significant effects on b-endorphin or cortisol concentrations related to medication type. There was no consistent relation between paradoxical responding to sedatives and opiate levels. An incidental finding of a two to three times higher incidence of paradoxical responding in the SIB plus stereotypy group compared to control groups confirmed the pattern of earlier results (Barron & Sandman, 1983, 1985).

Discussion

There are two general conclusions from these findings. First, patients with SIB or stereotypy had elevated b-endorphin levels compared with the retarded controls. This finding, coupled with the report of Gillberg et al. (1985) and the studies reporting effectiveness of opiate blockers in SIB, implicated b-endorphin in this treatment-resistant behavior. Because cortisol levels were not different, the significantly elevated b-endorphin levels in the SIB plus stereotypy group cannot be explained by general pituitary–adrenal dysfunction or increased stress levels. This finding is suggestive of either unique opiate receptor sensitivity in these patients or a dissociation between two POMC products. Although cortisol is not a direct POMC fragment, its release is stimulated by the POMC fragment, ACTH. The co-release of b-endorphin and ACTH is well-documented (Guillemin et al., 1977; Nakao, Jingami, Oki, Fukata, & Imura, 1979; Nakao, Oki, Tanaka, Horh, Nakai, Furui, Fukushima, Kuwayama, Kageyama, & Imura, 1980; Nakao, Nakai, Oki, Horii, & Imura, 1978) and was coupled in nonretarded subjects in the current study. A dissociation of b-endorphin and ACTH could result from specific enzyme activity either to cleave the peptides from POMC or to degrade the active fragment after it has been liberated. The biological program for directing the translational enzymes can follow either genetic predetermination or environmental demand (Farah, Millington, & O'Donohue, 1986). Thus, genetic or environmental factors may operate (or cooperate) in the availability of b-endorphin and the control of SIB.

The significance of this finding may be related to the two central hypotheses proposed to account for SIB. One hypothesis is that SIB reflects an insensitivity to pain and perhaps a general sensory depression (Barron & Sandman, 1984, 1986; Cataldo and Harris, 1982; Davidson et al., 1983; Deutsch, 1986; Farber, 1987; Sandman, 1988; Sandman et al., 1983). The strongest evidence in support of this hypothesis is attenuation of SIB with opiate blockers (Herman et al., 1987; Richardson & Zaleski, 1983; Sandman et al., 1983, 1988). Furthermore, opiate blockers reverse congenital insensitivity to pain (Dehen, Willer, Boureau, & Cambier, 1977; Pirodsky et al., 1985) and hypothalamic dysfunction coexisting with elevated pain threshold (Dunger, Leonard, Wolff, & Preece, 1980). These observations are consistent with an extensive animal literature demonstrating the hyperalgesic influence of naloxone (Grevert & Goldstein, 1977; Sandman et al., 1979). Thus, naloxone might attenuate SIB by increasing pain during the abusive act. Unfortunately, pain threshold has been difficult to assess in this population despite several attempts (Davidson et al., 1983; Richardson & Zaleski, 1983). The pain hypothesis suggests that SIB is related to opiate receptor supersensitivity, presumably due to restricted receptor plasticity or to decreased biosynthesis of b-endorphin. The small doses of naloxone (0.1 to 1.2 mg) sufficient to attenuate SIB may be construed as consistent with a supersensitive hypothesis.

However, it is plausible that naloxone attenuates SIB by blocking opiate-mediated euphorogenic effects. In this context, the proposed purpose of SIB is pain-induced release of opiates. An opiate "high" might be stimulated by SIB (pain releases b-endorphin). In this analysis SIB might be construed as an addiction because it is maintained to supply the "fix" for tolerant, down-regulated opiate receptors. This possibility is consistent with the results of the current study and supported by findings that repeated b-endorphin administration results in tolerance (Lal, 1975; Madden, Akil, Patrick, & Barchas, 1977), physical dependence (Wei & Loh, 1976) and euphoric-like effects (Belluzzi & Stein, 1977). Evidence supporting this possibility would require sampling plasma for b-endorphin immediately after an SIB episode and comparing it with behaviorally stable levels.

The second conclusion was that institutionalized mentally retarded patients have lower plasma b-endorphin concentration than do age-matched, nonretarded controls. Although the significance of this difference is uncertain, the findings that cortisol concentration was not different between institutionalized patients and healthy controls suggests that pituitary–adrenal functioning and stress tolerance levels do not account for the b-endorphin findings. The uncoupling of b-endorphin and cortisol among retarded patients but not nonretarded controls was observed in the correlational analyses; b-endorphin and cortisol levels were related significantly in nonretarded controls, especially in the morning samples. However, the correlation between these markers were not related in any of the patient groups. Thus, covariation of two biological markers related to stress, characterized the nonretarded group (cf. Guillemin et al., 1977; Nakao et al., 1978, 1979, 1980) but were clearly uncoupled in the patients. This pattern suggests that the differences between control subjects and patients is specific for b-endorphin and may relate to several obvious factors, such as institutionalization or the presence of mental retardation.

The results of this study may resolve an apparent contradiction in the literature. Wiezman et al. (1984) reported that autistic patients had lower plasma concentrations of humoral endorphin than did nonautistic control subjects. Gillberg et al. (1985) found that autistic patients with SIB had higher levels of b-endorphin than did patient control subjects. It could be argued that the disagreement between these findings may reflect differences between plasma and cerebrospinal fluid b-endorphin. Although b-endorphin concentration is higher in cerebrospinal fluid than in plasma, it is not determined whether cerebrospinal fluid and plasma b-endorphin are related (Naber et al., 1981; Nakao, Oki et al., 1980). Plasma b-endorphin should be vulnerable to episodic changes in enzyme activity and may not reflect brain concentrations. Thus, chronic differences in plasma b-endorphin among groups may suggest genetic disregulation of enzymes for its activation or degradation.

However, based on the current findings, it is plausible that the difference in the Wiezman et al. (1984) and Gillberg et al. (1985) studies is due to the choice of control groups. In the present study patients with the autistic-like behaviors of SIB and stereotypy had lower b-endorphin concentrations than did healthy controls (cf. Wiezman et al. 1984) but significantly higher b-endorphin than patient controls (cf. Gillberg et al., 1985). These results emphasize the importance of the comparison groups in studies of institutionalized patients. Clearly, opposite conclusions can be derived depending on which reference group is used.

These findings should be viewed as preliminary. There is considerable overlap in b-endorphin concentration among patient groups, perhaps reflecting the heterogeneous nature of SIB or the unreliability of b-endorphin as a marker. It is highly likely that b-endorphin is not centrally implicated in all patients with SIB (Barron & Sandman, 1985). For instance, not all patients with SIB respond favorably to opiate blockers. The variability of b-endorphin among patients may reflect the existence of subgroups of SIB patients whose behavior is controlled by b-endorphin disregulation. If subgroups exist, they may be candidates for specific treatments with agents such as opiate blockers.

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